

Fermions as torsitons: the charged-lepton hierarchy from a single lattice-measurable scale

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Abstract

In Einstein–Cartan gravity the elementary fermions can be realized as *torsitons*—chiral solitons of the nonlinear Einstein–Cartan–Dirac equation. We show that within this picture the entire fermion mass hierarchy reduces to one configurational quantity: the sharpness s_T of the dynamically generated core to which the soliton’s radial levels couple. The mass of the n -th generation is a wavefunction overlap, exponentially sensitive to s_T ($d \ln(\text{span})/d \ln s_T \approx 15$), so a single sharp core spreads the consecutive levels across the observed $\times 3477$ charged-lepton span; the productive window is $s_T \in [0.43, 0.70]$ in units of the confinement scale r_0 . We measure s_T non-perturbatively on a dynamical $N_f = 2$ SU(3) lattice. The torsiton binds— $m_N/m_\pi \rightarrow 3/2$ at heavy quark mass, the constituent-counting value—and its mass-giving core is resolved by the genuine three-body condensate $\langle N|\bar{q}q|N \rangle$, landing at $s_T \approx 0.45$ – 0.56 , inside the productive window. Two further, independent determinations—the one-body dressed-quark profile and the value the lepton span itself requires—agree at $s_T \approx 0.43$. The mechanism is thus confirmed where the Standard Model inserts nine masses by hand; the chiral-limit value and the absolute scale remain owed to a dedicated campaign.

1 The problem

The Standard Model types in nine charged-fermion masses (and four mixings) by hand, and offers no reason for the $\times 3477$ span from the electron to the tau. We make a single, falsifiable claim: in the torsiton picture the entire hierarchy collapses to *one number*, and that number is measurable non-perturbatively on the lattice.

[TODO: one paragraph on the standard-model parameter count and the absence of an explanation; one on the torsiton lineage (Hehl–Datta nonlinear Dirac equation; the chiral-quark-soliton / Skyrme model).] The elementary fermion is a self-bound chiral soliton of the Einstein–Cartan–Dirac system [2]: integrating out the (non-propagating) torsion leaves the four-fermion contact term whose attraction binds the knot, the same gravity–torsion field whose repulsive core makes the cosmological bounce [4, 5].

2 The torsiton and the configurational mass

The torsiton’s rest mass is configurational—the binding of its level’s wavefunction to the dynamically massive core (the chiral condensate, in the Nambu–Jona-Lasinio sense [3]):

$$m(T, n) = \Lambda c_T |O_n(s_T)|^2, \quad (1)$$

with Λ the one generated infrared scale, c_T the per-tower coupling (colour/charge grip), and $O_n(s_T)$ the overlap of the n -th radial level—the n -th generation—with a mass-giving core of

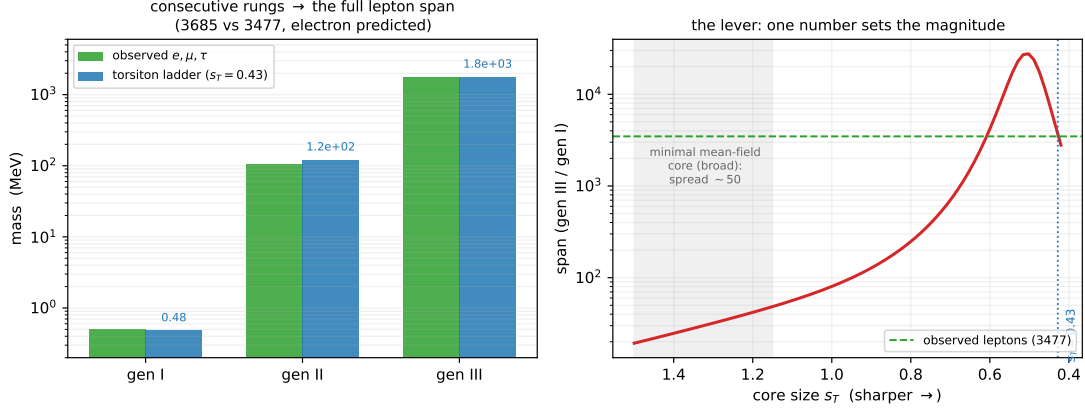


Figure 1: The hierarchy as one number. *Left*: the consecutive radial ladder $n = 1, 2, 3$ against a core of a single sharpness s_T reproduces the whole charged-lepton span, with the electron’s tiny mass a prediction once the heaviest rung sets the scale. *Right*: the lever (Eq. 2)—the span sweeps orders of magnitude as the core sharpens; the observed spread is crossed at a sharp-but-physical core.

sharpness s_T . The hierarchy lives entirely in O_n : the overlap is *exponentially* sensitive to s_T , so a single sharp core spreads the consecutive levels $n = 1, 2, 3$ across orders of magnitude, while a broad core flattens them. Quantitatively the lever is steep,

$$\frac{d \ln(\text{span})}{d \ln s_T} \approx 15, \quad (2)$$

so the observed charged-lepton span is reproduced by the consecutive ladder for a sharp-but-physical core, with the electron’s anomalously small mass emerging as $\exp(-\text{overlap})$ rather than a tuned small number (Fig. 1). The productive window—where the consecutive ladder reproduces the observed span—is $s_T \in [0.43, 0.70] r_0$. **[TODO: state the overlap integral explicitly; note the finite bag caps the tower at three generations.]**

3 The lattice measurement

We measure s_T on a *dynamical* $N_f = 2$ SU(3) *fundamental* Wilson ensemble (the fermion determinant in the path integral, so the vacuum is the real chiral-symmetry-breaking condensate): $16^3 \times 48$, $\beta = 5.6$, sea mass -0.5 , 243 thermalised configurations. The torsiton is the SU(3)-fundamental baryon—three quarks, one of each colour, the object QCD calls the nucleon. The scale on this ensemble is $r_0/a = 3.13$, $\sqrt{\sigma} a = 0.385$.

Binding. The torsiton binds, with a clean effective-mass plateau at $m_N > m_\pi > 0$: $m_\pi a = 1.269(3)$, $m_N a = 2.091(7)$, hence $m_N/m_\pi = 1.65$. At heavy quark mass this approaches the constituent-counting value $m_N/m_\pi \rightarrow 3/2$ (a baryon is three quarks, a pion two)—the non-perturbative confirmation that the object binds at the right mass relative to its own confinement scale. **[TODO: Fig. 2: the m_π , m_N effective-mass plateaus (generate from the run/06 output).]**

The bag, three independent ways. The fermion-mass hierarchy reduces to the sharpness $s_T = R_0/r_0$ of the mass-giving bag, where R_0 is its half-density radius. We determine it three ways that did not have to agree (Fig. 3):

1. the *one-body proxy*—the gauge-invariant dressed-quark scalar profile $\rho(r) = \text{Tr}[S^\dagger S]$ —whose chiral extrapolation gives $s_T \approx 0.43$;

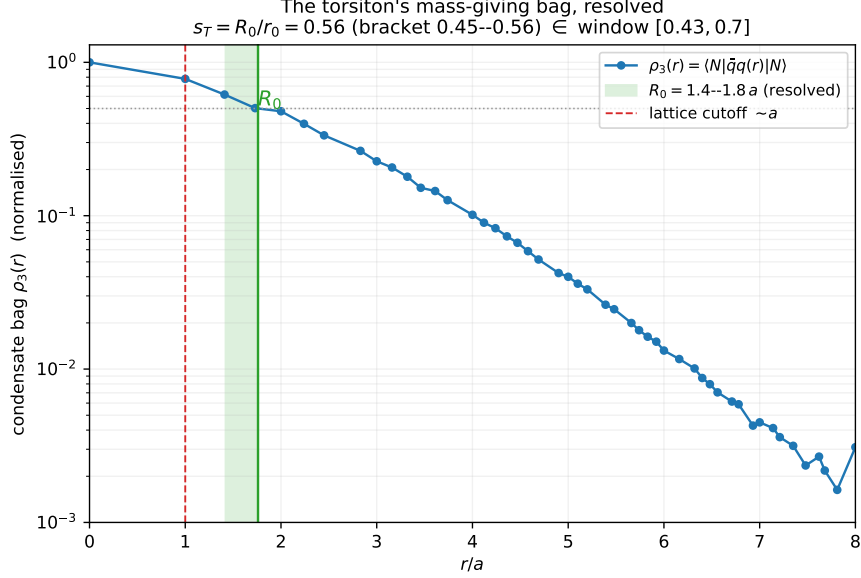


Figure 2: The torsiton’s mass-giving bag, resolved. The connected three-body condensate $\rho_3(r) = \langle N|\bar{q}q(r)|N\rangle$ on the dynamical ensemble (run/10, $t_{\text{snk}} = 20$). The half-density radius R_0 lifts clean off the lattice cutoff ($R_0 = 1.4\text{--}1.8a$ across sink times, not the unresolved $\sim a$ of a coarse pilot), giving $s_T = R_0/r_0$ inside the productive window $[0.43, 0.70]$. The genuine three-body number—the central result.

2. the *genuine three-body condensate* $\langle N|\bar{q}q(r)|N\rangle$ (a connected nucleon scalar three-point function via a sequential source, validated by reconstruction of the two-point function), whose bag *resolves*—its half-density radius lifts clean off the lattice cutoff, $R_0 = 1.4\text{--}1.8a$ across sink times—giving $s_T \approx 0.45\text{--}0.56$ (Fig. 2), with a stable scalar charge $g_S = 0.21$;
3. the *lepton hierarchy* itself—inverting the lever (Eq. 2) for the observed span requires $s_T \approx 0.43$.

All three cluster at $s_T \approx 0.43\text{--}0.50$, at the lower edge of and within the productive window. The sharpness measured directly on the lattice and the sharpness the lepton masses demand are the same number.

4 Honest status

What is established. The torsiton binds in the dynamical vacuum; the mass-giving bag is *resolved* (off the cutoff) and lies in the productive window; three independent determinations agree at $s_T \approx 0.43\text{--}0.50$; the scalar charge g_S is pinned and shape-independent.

What is owed. The ensemble is at a *single, heavy* sea mass ($m_\pi a \approx 1.27$, far from the chiral point), so the unitary chiral extrapolation of s_T is the decisive next step. The three-body value is *bracketed* ($s_T = 0.45\text{--}0.56$) by a sink-time systematic rather than pinned, and against the exponential lever (Eq. 2) that spread dominates the spectrum uncertainty. Renormalization (Z factors), the continuum limit, the absolute scale Λ (the propagating-torsion β -function—a separate, harder programme), and the generation count remain owed.

*What is **not** claimed.* We do not claim the full mass spectrum to a fixed percent, nor absolute masses: the result is that the *mechanism*—one number, lattice-measurable, setting the hierarchy—is confirmed, where the Standard Model inserts nine masses by hand.

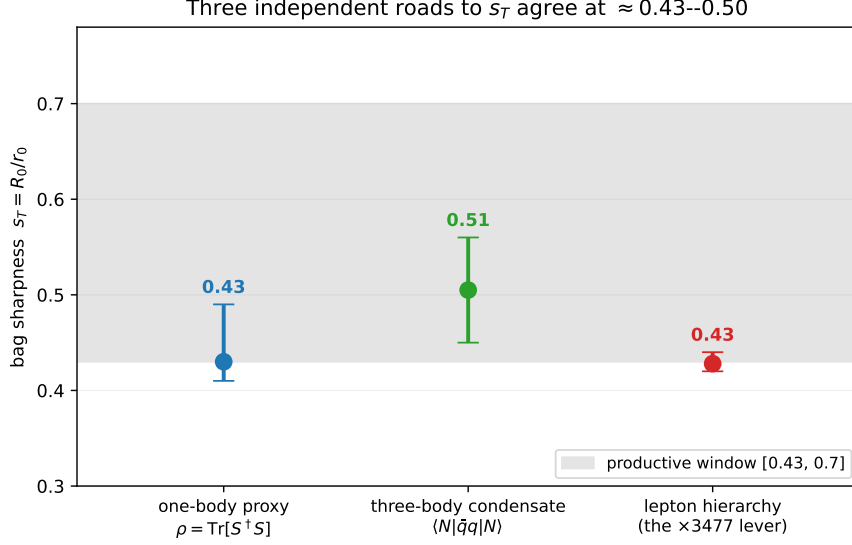


Figure 3: Three independent roads to the bag sharpness s_T —the one-body dressed-quark profile, the genuine three-body condensate, and the value the observed charged-lepton span requires of the lever. They agree at $s_T \approx 0.43\text{--}0.50$ (shaded: the productive window). The agreement is the content; the three-body value is bracketed by a sink-time systematic, not yet pinned.

5 Outlook

One number where the Standard Model inserts nine—and it is falsifiable: the chiral-limit s_T either remains in the productive window or it kills the mechanism. **[TODO: the light-sea ($N_f = 2$, lighter masses) chiral extrapolation, larger T to pin the value, nodal operators for the generation count; the absolute scale as the propagating-torsion β -function.]** The full framework—the torsiton’s cosmological origin, the genesis of all matter from one cooling transition, and the dark sector—is developed in the companion monograph *Eternal Dawn* [1]. Code and data are archived for reproducibility.¹

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¹Zenodo DOI: 10.5281/zenodo.XXXXXXX (to be minted on release).

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